

Bayesian Bandit Encoding: Analysis on a Two-State Markov Channel

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Abstract—We introduce Bayesian bandit encoding (BBE), a framework that selects how to encode packets for a channel whose parameters are unknown. BBE has two components. A Bayesian learner maintains posteriors over the quantities that determine performance, updated from the noise patterns recovered by a guessing random additive noise decoding (GRAND) receiver and from packet acknowledgments. The transmitter selects an encoding strategy by Thompson sampling from these posteriors, and the receiver uses the learned channel parameters to order its noise guesses. We analyze BBE on a two-state Markov burst channel, where the transmitter decides whether to interleave a random linear code. For this channel, the GRAND query order has a closed form, the block error rate has a closed-form expression under an independent false-hit model, and interleaving to depth D shrinks the channel's memory from λ to λ^D . Interleaving helps a decoder that ignores the bursts more than one that has learned them, so the better encoding can change as the receiver learns. When every frame reveals its full noise sequence, BBE's regret is bounded uniformly in time under a margin condition. In practice GRAND reveals the noise only of codewords it decodes, so the encoding choice determines what the receiver learns. Simulations show that the BBE variants whose selection also uses acknowledgments interleave early when the initial belief is that the channel is memoryless, learn the channel memory faster, and achieve lower regret than selection from noise patterns alone.

I. INTRODUCTION

Encoders and decoders are usually designed for an assumed channel. Interleaving is a standard example. A block interleaver spreads each burst of errors over many codewords, so that a decoder designed for independent errors can correct them, and it pays for this with delay. A decoder that knows the burst statistics can exploit the bursts instead, and for such a decoder the interleaver discards useful structure [1]. The better encoding therefore depends on the channel and on how much the receiver knows about it, and neither is known when a link starts.

We introduce Bayesian bandit encoding (BBE), which learns the channel while the link operates and uses what it learns at both ends. A Bayesian learner maintains a posterior over the quantities that determine performance. Its main evidence comes from the receiver, which runs guessing random additive noise decoding (GRAND) [2]. GRAND guesses noise patterns in decreasing order of likelihood and stops at the first one that turns the received word into a codeword, so every decoded codeword also reveals the noise the channel applied. Packet acknowledgments provide further evidence. The

transmitter selects the encoding of each frame by Thompson sampling from the posterior, and the receiver orders its guesses by the learned channel parameters. Encoder and decoder are adapted from the channel and action posteriors in a single loop. Two unpublished manuscripts demonstrated BBE in simulation, on a channel with several on-off interferers [3] and with catalogs of more than a thousand code and decoder configurations [4]. This paper gives the first analysis.

We study BBE on a two-state Markov burst channel, where the transmitter decides in each frame whether to send a random linear code (RLC) plain or through a block interleaver. On this channel the GRAND query order and the posterior update have closed forms, the block error rate (BLER) has a closed-form expression under an independent false-hit model, and the instance exhibits the effect that motivates BBE: the best encoding changes as the receiver learns. At our operating point, interleaving lowers the BLER of a memoryless decoder from 0.089 to 0.039. Once the decoder has learned the bursts, plain transmission reaches 0.0097, nine times lower than at the start and three times lower than interleaving. BBE finds this switch on its own. Our contributions are:

- *The BBE framework.* Bayesian learning from decoded noise and acknowledgments, Thompson-sampling selection of the encoding, and a GRAND query order driven by the learned channel (Section II), with the explicit update and sampling rules for the two-state channel (Section IV).
- *Analysis of GRAND on the two-state channel.* The 2^n noise patterns fall into $O(n^2)$ classes of equally likely patterns, which gives the query order and a closed-form BLER under an independent false-hit model, validated against an explicit RLC decoder, and an interleaved codeword sees a two-state channel with memory λ^D , so the better encoding can change as the receiver learns (Section III).
- *Bounded regret.* When every frame reveals its noise sequence, the regret of BBE is bounded uniformly in time, with explicit constants in the channel, the query budget and the penalty (Theorem 1).
- *Closed-loop simulations.* All BBE variants learn the bursts and have a fraction of the regret of a conventional interleaved link, and BBE-ACK and BBE-hybrid interleave early and move to plain transmission within the first tens of frames (Section VI).

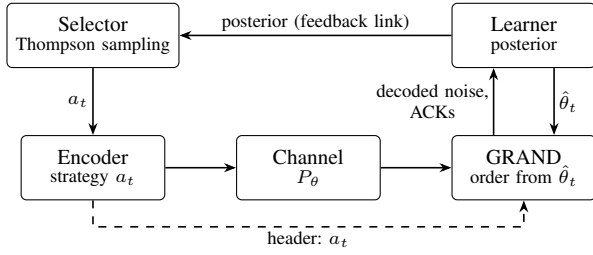


Fig. 1. One frame of BBE.

Algorithm 1 One frame of Bayesian bandit encoding

- 1: Receiver sets its GRAND order from the posterior mean $\hat{\theta}_t$
- 2: Transmitter draws a sample from the posterior, plays the action a_t with the lowest sampled cost, and announces a_t in a header
- 3: GRAND decodes each codeword in the order implied by $\hat{\theta}_t$
- 4: Decoded noise updates the channel posterior; ACKs update the per-action posteriors

GRAND has been extended to soft information [6], [7] and to channels with memory. With a known Markov model it outperforms interleaving [1], and universal variants estimate an unknown finite-state channel inside the decoder [8]. In BBE the learned channel also drives the transmitter's choice of encoding. Bandit algorithms have been applied to rate adaptation [9], [10], and bandits whose arms share a common parameter can have bounded regret [11], [12]. In BBE the channel is that shared parameter, and Theorem 1 gives the constants for the two-state channel.

II. BAYESIAN BANDIT ENCODING

The channel belongs to a parametric family $\{P_\theta\}$. In each frame the transmitter selects an action a from a finite set $\mathcal{A}$. An action fixes the encoding strategy and may also fix decoder parameters such as the GRAND query budget. A *performance model* gives the expected cost $J_a(\theta; \hat{\theta})$ of action a when the channel is θ and the receiver orders its queries by the noise model $P_{\hat{\theta}}$. Fig. 1 and Algorithm 1 describe one frame.

The learner holds two kinds of posterior. The *channel posterior* over θ is updated from decoded noise, and its mean $\hat{\theta}_t$ configures the decoder. The *action posteriors* are Beta distributions on each action's codeword loss rate, updated from acknowledgments. The selector can draw from either or both, which gives three instances of BBE:

- *BBE-noise* draws $\tilde{\theta}_t$ from the channel posterior and plays $\arg \min_a J_a(\tilde{\theta}_t; \hat{\theta}_t)$. Every action is scored through one channel model, so an observation made under one action updates the predicted cost of all.
- *BBE-ACK* draws a loss rate for each action from its action posterior and adds the delay penalty. It measures each action's performance directly.

TABLE I
NOTATION

$\theta = (p, q)$	transition probabilities $\mathbb{P}(1 0)$, $\mathbb{P}(1 1)$ of the noise
π_1, λ	error rate $p/(1 - q + p)$, memory $q - p$
n, k, r	code length, dimension, redundancy $n - k$
D, M	interleaver depth (codewords per frame), $M = Dn$
A, B	plain and interleaved transmission
Q	GRAND query budget per codeword
c	delay penalty for interleaving
$J_a(\theta; \hat{\theta})$	expected cost of a on channel θ , decoder belief $\hat{\theta}$
$\hat{\theta}_t, \tilde{\theta}_t$	posterior mean and posterior sample in frame t
$\hat{\pi}_1$	initial estimate of the error rate
κ	prior strength: pseudo-transitions per parameter

- *BBE-hybrid* uses the action posteriors with the channel model's prediction $\text{BLER}_a(\hat{\theta}_t; \hat{\theta}_t)$ added as prior pseudo-counts.

In all three, the decoder is configured from the channel posterior, and Section IV gives the explicit rules for the two-state channel. BBE requires a posterior update from decoded noise and a performance model. For the two-state channel the update is conjugate and the performance model has a closed form, and Sections III and IV state where each is approximate. For other channels they can be approximated or estimated by simulation, as in [3].

III. THE TWO-STATE MARKOV INSTANCE

A. Channel, code and strategies

The channel flips bit t when $Z_t = 1$, where $Z_1, Z_2, \dots$ is a stationary two-state Markov chain with $p = \mathbb{P}(Z_{t+1} = 1 | Z_t = 0)$ and $q = \mathbb{P}(Z_{t+1} = 1 | Z_t = 1)$. Table I collects the notation. Bursts last $1/(1 - q)$ bits on average, and $p = q$ gives a memoryless channel.

A frame carries D codewords of a binary (n, k) RLC. Strategy A sends them one after another. Strategy B writes them as the rows of a $D \times n$ array and sends the array column by column, so codeword j occupies channel uses $j, j + D, j + 2D, \dots$. The cost of a frame is the fraction of codewords lost plus the delay penalty,

$$J_a(\theta; \hat{\theta}) = \text{BLER}_a(\theta; \hat{\theta}) + c \mathbf{1}\{a = \text{B}\}, \quad (1)$$

and $J^* = \min_a J_a(\theta; \theta)$ is the optimal cost with a matched decoder.

Lemma 1. *The noise seen by one codeword under B is a stationary two-state Markov chain with error rate π_1 and memory λ^D , that is, with parameters $\theta^{(D)} = (\pi_1(1 - \lambda^D), \pi_1 + \pi_0 \lambda^D)$.*

Proof. With $\Pi = \mathbf{1}\pi^\top$, the transition matrix is $P = \Pi + \lambda(I - \Pi)$. The matrices Π and $I - \Pi$ are complementary projections, so $P^D = \Pi + \lambda^D(I - \Pi)$. $\square$

The receiver decodes strategy B by applying the same map to its belief, $\hat{\theta} \mapsto \hat{\theta}^{(D)}$.

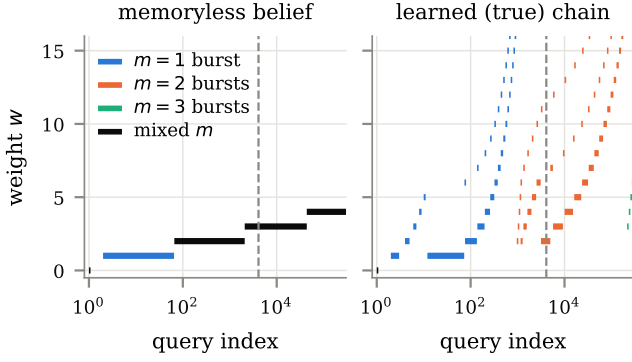


Fig. 2. GRAND query order for $n = 64$ under a memoryless belief (left) and under the learned channel, $\pi_1 = 0.01$ with mean burst length 4 (right). Each segment is a tie group, placed at its weight over the query indices it occupies and colored by its number of bursts m . The dashed line is the budget $Q = 2^{12}$.

B. GRAND on the two-state channel

GRAND with abandonment [2] queries noise patterns in decreasing order of probability under its belief and returns the first pattern z for which $y \oplus z$ is a codeword. It declares an erasure after Q queries. It errs if the true pattern lies beyond the budget or if an earlier query passes the parity check, which happens with probability 2^{-r} per query for an RLC.

For a pattern $z \in \{0, 1\}^n$, let w be its weight, m its number of bursts (maximal runs of ones), $s = z_1$ and $e = z_n$.

Lemma 2. $\mathbb{P}_\theta(Z = z) = \pi_s(1-p)^{n_{00}}p^{n_{01}}(1-q)^{n_{10}}q^{n_{11}}$, where $n_{01} = m - s$, $n_{10} = m - e$, $n_{11} = w - m$ and $n_{00} = n - 1 - w - m + s + e$. The probability therefore depends on z only through (w, m, s, e) .

Proof. Each burst not starting at position 1 is entered by a $0 \rightarrow 1$ transition, and each burst not ending at position n is left by a $1 \rightarrow 0$ transition. $\square$

Patterns with the same weight, number of bursts and end-points are equally likely, so the 2^n patterns fall into $O(n^2)$ classes, and a GRAND query order is a sort of these classes. The size of each class is a product of two binomial coefficients, derived in the extended version [5]. Classes that tie under the belief form a tie group, whose members we query in random order. For patterns with $s = e = 0$, Lemma 2 gives $-\log \mathbb{P}_\theta(z) = \text{const} + m\beta + (w - m)\gamma$ with $\beta = \log \frac{(1-p)^2}{p(1-q)}$ and $\gamma = \log \frac{1-p}{q}$: each new burst costs β and each further error within a burst costs γ . When $\lambda > 0$, $\beta > \gamma$ and GRAND prefers few long bursts to many scattered errors. When $p = q$ the order reduces to the Hamming-weight order of classical GRAND. Fig. 2 shows the effect. Within 2^{12} queries the learned order covers every single burst of up to 31 errors, while the Hamming order covers the patterns of weight at most two and about 5% of those of weight three.

Proposition 1. Sort the tie groups of $\hat{\theta}$ by decreasing likelihood, let A_g be the number of patterns in groups ranked before

group g , and let $\varepsilon = 2^{-r}$. If each incorrect query passes the parity check independently with probability ε , then

$$1 - \text{BLER}(\theta; \hat{\theta}) = \sum_g \frac{\mathbb{P}_\theta(Z \in g)}{|g|} \sum_{j=1}^{L_g} (1 - \varepsilon)^{A_g + j - 1}, \quad (2)$$

where $L_g = \min\{|g|, (Q - A_g)^+\}$ is the number of members of g within the budget.

Proof. Because queries within each tie group are uniformly randomly ordered, every pattern in g has a uniform rank within that group, so given $Z \in g$ its rank is $A_g + U$ with U uniform on $\{1, \dots, |g|\}$. Decoding succeeds iff $A_g + U \leq Q$ and none of the $A_g + U - 1$ earlier queries is a false hit. $\square$

Equation (2) is the performance model of BBE for this channel. Fig. 3 compares (2) with a real GRAND decoder on an explicit RLC(64, 44) (2×10^4 blocks per point); they agree within Monte Carlo error at all 16 points. The figure also shows why the encoder and the decoder must be adapted together. With the memoryless decoder, interleaving has the lower BLER at every budget above 2^7 . With the learned decoder, plain transmission has the lower BLER above 2^{10} , and at $Q = 2^{12}$ it is nine times lower than with the memoryless decoder.

The reason is that interleaving helps a memoryless decoder more than a matched decoder at this operating point. A memoryless decoder queries by weight, so it decodes a codeword when the number of errors in it is small. Interleaving leaves the mean number of errors per codeword unchanged but spreads each burst over D codewords, so fewer codewords receive many errors. A decoder that knows the bursts already queries long single bursts early, and for it interleaving mainly makes each codeword's noise less predictable, while adding delay. With decoder belief $\hat{\theta}$, interleaving has the lower cost iff c is below the critical penalty $c^*(\theta, \hat{\theta}) = \text{BLER}_A(\theta; \hat{\theta}) - \text{BLER}_B(\theta; \hat{\theta})$. Writing c_{mem}^* and c_{mat}^* for the critical penalties of the memoryless and the matched decoder, the encoding has to switch when

$$c_{\text{mat}}^* < c < c_{\text{mem}}^*. \quad (3)$$

In this interval interleaving is cheaper with the initial memoryless decoder and plain transmission is cheaper with the matched decoder. This is the switch that BBE has to find.

IV. BBE ON THE TWO-STATE CHANNEL

We now spell out each step of Algorithm 1 for the two-state channel.

A. What a frame reveals

When GRAND decodes a codeword correctly, the pattern it returns is the noise on that codeword's channel uses. An earlier false hit, with probability at most $Q2^{-r}$, returns a wrong pattern; acknowledgments count such codewords as losses, as an outer error-detecting code would, which could also exclude them from the channel update. The receiver writes the decoded patterns back into channel order and marks the bits of lost

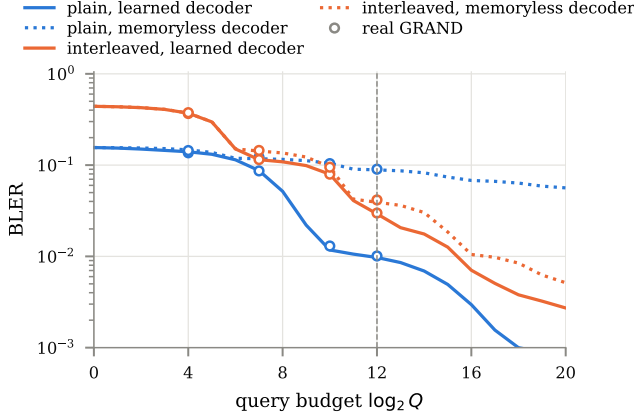


Fig. 3. BLER versus query budget for RLC(64, 44), $D = 8$, on the channel of Fig. 2: plain and interleaved transmission, each with the memoryless and the learned decoder. Lines: (2). Markers: a real GRAND decoder. The dashed line marks $Q = 2^{12}$.

codewords as unknown. It then counts, over adjacent channel uses $(t, t+1)$ whose bits are both known, the transitions

$$n_{ij} = \#\{t : Z_t = i, Z_{t+1} = j\}, \quad i, j \in \{0, 1\}. \quad (4)$$

For example, the known segment 0011100 gives $n_{00} = 2$, $n_{01} = 1$, $n_{11} = 2$ and $n_{10} = 1$. Under strategy A a decoded codeword covers n consecutive channel uses and contributes $n-1$ transitions. Under B adjacent channel uses belong to consecutive codewords, so a transition is known when both codewords are decoded, and a fully decoded frame yields all $M-1$ transitions.

The two parameters are learned at very different speeds. Transitions out of state i occur about $\pi_i M$ times per frame. Errors are rare, so $\pi_1 \ll \pi_0$, and the burst persistence q is learned about π_0/π_1 times more slowly than p .

B. Bayesian update

Conditional on the first bit of each run of known bits, the likelihood of the counts is $p^{n_{01}}(1-p)^{n_{00}} \cdot q^{n_{11}}(1-q)^{n_{10}}$, a product of two Bernoulli likelihoods, one for each current state. Independent Beta priors are therefore conjugate. With $p \sim \text{Beta}(\alpha_0, \beta_0)$ and $q \sim \text{Beta}(\alpha_1, \beta_1)$, the update after a frame is

$$\alpha_0 \leftarrow \alpha_0 + n_{01}, \quad \beta_0 \leftarrow \beta_0 + n_{00}, \quad \alpha_1 \leftarrow \alpha_1 + n_{11}, \quad \beta_1 \leftarrow \beta_1 + n_{10}, \quad (5)$$

and the posterior mean is $\hat{\theta} = (\hat{p}, \hat{q})$ with $\hat{p} = \alpha_0/(\alpha_0 + \beta_0)$ and $\hat{q} = \alpha_1/(\alpha_1 + \beta_1)$. The whole posterior is four numbers, and the receiver can return it to the transmitter in a few bytes per frame. Under full feedback this is the exact posterior, given the first bit. Under decoded feedback, which transitions are observed depends on the noise through the decoding outcome; the update ignores this observation mechanism and therefore defines an approximate posterior. Section V-B analyzes the resulting estimation bias.

The initial knowledge is an estimate $\hat{\pi}_1$ of the error rate, with nothing known about memory. The prior sets $\alpha_i = \kappa \hat{\pi}_1$

and $\beta_i = \kappa(1 - \hat{\pi}_1)$ for $i = 0, 1$, so both parameters start at $\hat{\pi}_1$: a memoryless channel. The prior strength κ counts pseudo-transitions. After N_i observed transitions out of state i , the prior weighs $\kappa/(\kappa + N_i)$ in the posterior mean, and the posterior standard deviation of θ_i is about $\sqrt{\theta_i(1-\theta_i)/(\kappa + N_i)}$. The prior on θ_i thus carries the weight of about $\kappa/(\pi_i M)$ frames of observations, which is small for p and large for q . Before each frame the receiver sorts the pattern classes by their log-probability under $\hat{\theta}$ (under $\hat{\theta}^{(D)}$ for B). At the prior this is Hamming-weight GRAND, and as $\hat{q}$ grows, long single bursts move ahead of scattered errors.

C. Selecting the encoding

The cost (1) combines the BLER with the delay penalty. The three variants estimate the BLER differently.

BBE-noise draws $\tilde{p} \sim \text{Beta}(\alpha_0, \beta_0)$ and $\tilde{q} \sim \text{Beta}(\alpha_1, \beta_1)$, and evaluates $J_a(\tilde{\theta}; \hat{\theta})$ with (2). In this evaluation the order of the tie groups, and hence A_g and L_g , comes from the decoder belief $\hat{\theta}$, and the group probabilities $\mathbb{P}_{\hat{\theta}}(Z \in g)$ come from the sample $\tilde{\theta}$ (both mapped by Lemma 1 for B). It plays the strategy with the lower cost. The greedy version uses $\tilde{\theta} = \hat{\theta}$.

BBE-ACK keeps, for each strategy a , discounted counts L_a and S_a of lost and decoded codewords. After a frame with strategy a_t and ℓ_t of D codewords lost,

$$L_a \leftarrow \gamma L_a + \ell_t \mathbf{1}\{a = a_t\}, \quad S_a \leftarrow \gamma S_a + (D - \ell_t) \mathbf{1}\{a = a_t\}, \quad (6)$$

for both a , with discount $\gamma < 1$, because the loss rate of each strategy falls as the decoder learns. It draws $\tilde{\epsilon}_a \sim \text{Beta}(1 + L_a, 1 + S_a)$ and plays $\arg \min_a \tilde{\epsilon}_a + c \mathbf{1}\{a = B\}$.

BBE-hybrid draws $\tilde{\theta}$ as *BBE-noise* does, predicts $b_a = \text{BLER}_a(\tilde{\theta}; \hat{\theta})$, and adds m_0 pseudo-codewords to the counts of *BBE-ACK*: $\tilde{\epsilon}_a \sim \text{Beta}(1 + L_a + m_0 b_a, 1 + S_a + m_0(1 - b_a))$.

V. ANALYSIS

A. Bounded regret under full feedback

We measure BBE against a system that knows θ by the regret $R_T = \sum_{t \leq T} \mathbb{E}[J_{a_t}(\theta; \hat{\theta}_t)] - TJ^*$, and analyze BBE-noise under *full feedback*, in which the learner sees the noise sequence of every frame. The result needs a margin condition on the neighborhood $B = \{(p', q') : |p' - p| < \rho_0, |q' - q| < \rho_1\}$ of the true parameters.

Assumption 1. For some $\rho_0, \rho_1 \in (0, 1]$: (a) every decoder belief $\hat{\theta} \in B$ gives the matched query order through the budget, including tie handling, for both strategies; and (b) for every decoder belief $\hat{\theta} \in B$ and sampled parameter $\theta \in B$, the strategy that minimizes $J_a(\hat{\theta}; \theta)$ is the optimal one.

Condition (a) holds near almost every θ , because the query order sorts finitely many pattern probabilities that vary continuously with the belief. For a small enough neighborhood, (b) then holds whenever c differs from the matched critical penalty. The extended version [5] gives the precise conditions.

The argument behind the theorem is simple. Once the posterior mean and the posterior sample both lie in B , the decoder uses the matched order and the sample ranks the

two strategies correctly, so the frame costs J^* . Under full feedback each frame adds about $\pi_i M$ transitions out of state i whatever strategy is played, so the posterior concentrates at a rate that does not depend on the selector's choices. The expected number of frames spent outside B is therefore finite, and it does not grow with the horizon.

Theorem 1. *Assume full feedback and Assumption 1, and let $d_0 = |\hat{\pi}_1 - p|$, $d_1 = |\hat{\pi}_1 - q|$ and $\kappa \geq 1$. BBE-noise, and its greedy variant with $\theta_t = \hat{\theta}_t$, satisfy for every T*

$$R_T \leq (1+c) \left[K_0 + \sum_{i \in \{0,1\}} \frac{1}{\pi_i M} \left(\frac{6\kappa d_i}{\rho_i} + \frac{45}{\rho_i^2} \right) \right], \quad (7)$$

where K_0 depends on θ , κ and M but not on the margin.

Proof sketch. Full feedback supplies about $\pi_i M$ transitions out of state i in every frame, whichever strategy is played. After an initial period set by the prior strength κ , the probability that the posterior mean or the posterior sample lies outside B in frame t therefore decays exponentially in t , and a frame outside B costs at most $1+c$ more than J^* . Summing these probabilities over t gives a finite bound. The extended version [5] gives the concentration inequalities and the constants. $\square$

B. Decoded feedback

In practice the learner sees the noise of decoded codewords only, and decoding fails most often on patterns the current decoder ranks low. Let $\mathcal{S}$ be the event that a plain codeword is decoded, ε_{dec} its BLER, and N_1 and N_{11} its counts of transitions out of state 1 and from 1 to 1. Counts from decoded codewords estimate $q_c = \mathbb{E}[N_{11} \mathbf{1}_{\mathcal{S}}] / \mathbb{E}[N_1 \mathbf{1}_{\mathcal{S}}]$, and a martingale argument in the extended version [5] gives

$$|q_c - q| \leq \sqrt{q(1-q)(n-1)\pi_1 \varepsilon_{\text{dec}}} / \mathbb{E}[N_1 \mathbf{1}_{\mathcal{S}}]. \quad (8)$$

The numerator grows with the decoding failures, through ε_{dec} , and the denominator is the number of error-state transitions that decoded codewords supply. The bias is therefore small when failures are rare relative to the observed error-state data, so learning and decoding improve together. For a learner that sends only plain frames, the long-run estimate is a fixed point of the map from a decoder belief to the estimate q_c its decoded codewords produce, and it can be computed by iterating that map. The encoding determines how fast the estimate approaches it. Interleaved frames decode many of the bursty codewords that plain frames lose, and after de-interleaving these codewords return the burst statistics the learner needs.

VI. SIMULATION RESULTS

A. Setup

We use an RLC(64, 44), $D = 8$ codewords per frame, $Q = 2^{12}$ and $c = 0.02$, on a channel with $\pi_1 = 0.01$ and mean burst length 4 ($p \approx 0.0025$, $q = 0.75$). The prior uses $\hat{\pi}_1 = \pi_1$ and $\kappa = 200$, which gives the prior on q the weight of about 40 frames of observations of the error state. Each run lasts

400 frames, and results are averages over 20 runs. The plotted costs are the expected costs (2) of each decision. Including the penalty, interleaving is the cheaper encoding for the initial decoder (0.059 against 0.089) and plain transmission for the matched one (0.0097 against 0.049), so $J^* = 0.0097$. The critical penalties are $c_{\text{mem}}^* = 0.0495$ and $c_{\text{mat}}^* = -0.0195$, so $c = 0.02$ lies in the range where the better encoding changes as the decoder learns. Each frame contains about 507 transitions out of state 0 and 5 out of state 1.

BBE-ACK and BBE-hybrid use the discount $\gamma = 0.95$, and BBE-hybrid gives the model's prediction the weight of $m_0 = 20$ codewords. We compare with three fixed designs: plain and interleaved transmission with the learning decoder, and a conventional link that interleaves every frame and keeps the memoryless decoder.

B. Results

Fig. 4 shows BBE in operation. As the learner picks up the bursts (panel (a)), the decoder improves and the cost of plain transmission falls from 0.089 toward J^* (panel (b)). It drops below the cost of interleaving after a median of 12 frames, close to the 8 frames predicted for full feedback by replacing the transition counts with their expected values.

BBE-ACK and BBE-hybrid follow this change (panel (c)). They interleave about half of the first 50 frames, while plain frames are still losing codewords, and then move to plain transmission. The interleaved frames speed up learning: after 50 frames the posterior mean of q is 0.29 under BBE-ACK and 0.30 under BBE-hybrid, against 0.17 under BBE-noise. BBE-noise sends plain from the first frame, although interleaving is the better choice for the initial decoder on the true, bursty channel. BBE-noise compares the encodings on channels drawn from its posterior, which early on have almost no memory, and on such channels interleaving changes nothing and the penalty favors plain. By the time the posterior has learned the memory, the decoder has too, and plain is then better on the true channel as well. (Where $c < c_{\text{mat}}^*$, BBE-noise interleaves once it has learned the memory.) Learning from plain frames alone loses little in the long run: the fixed point of Section V-B is $q_c = 0.749$ against $q = 0.75$, with the matched BLER to four digits.

Panel (d) and Table II compare regret. All three BBE variants are far below the fixed interleaved designs. Over 400 frames the conventional link accumulates 19.8, while BBE-ACK accumulates 3.91 and BBE-hybrid 4.22, and BBE-ACK's average cost per frame is 0.020 against 0.059. With full feedback, BBE-noise accumulates 1.72 by frame 400 and only 0.10 more over the next 1100 frames, as Theorem 1 predicts. The prior strength κ determines which evidence matters most. With a weak prior ($\kappa = 20$) the noise patterns alone override the memoryless belief within a few frames, and BBE-noise has the lowest regret among the BBE variants. With $\kappa = 200$ and $\kappa = 1000$, acknowledgments reveal early that plain frames lose more codewords than the prior predicts, and BBE-ACK and BBE-hybrid have 13–22% less regret than BBE-noise. Forcing interleaving for the first 10 frames, about the predicted

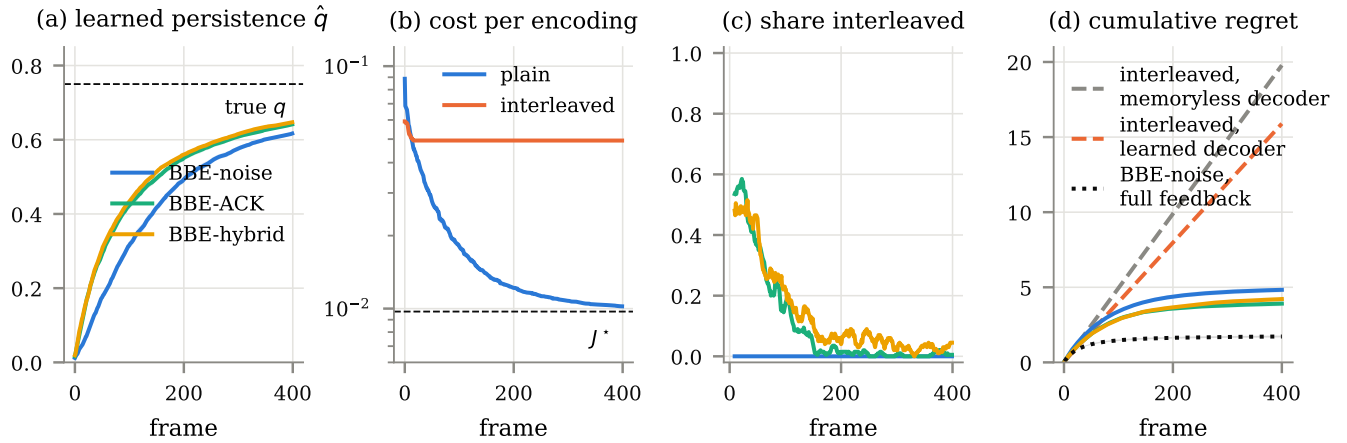


Fig. 4. BBE at the operating point ($\kappa = 200$, 20 runs, bands show one standard error). (a) Posterior mean of the burst persistence q . (b) Expected cost of each encoding with the receiver’s current decoder, in BBE-hybrid runs; the dashed line is J^* . (c) Fraction of runs interleaving (10-frame moving average). (d) Cumulative regret of the BBE variants (colors as in (a)) and of fixed designs.

TABLE II
TOTAL REGRET OVER 400 FRAMES (MEAN OF 20 RUNS).

	$\kappa = 20$	$\kappa = 200$	$\kappa = 1000$
BBE-noise	0.71	4.83	14.72
BBE-ACK	1.25	3.91	11.52
BBE-hybrid	1.16	4.22	11.81
plain, learned decoder	0.68	4.87	14.65
interleaved, learned decoder	15.85	15.89	16.02
interleaved, memoryless decoder		19.78	

switching time, also lowers BBE-noise’s regret at $\kappa = 200$, to 3.25. A log-normal error with $\sigma = 0.5$ in $\hat{\pi}_1$ changed no variant’s regret by more than one standard error.

VII. CONCLUSION

This paper introduced Bayesian bandit encoding, a framework in which a Bayesian learner fed by the noise patterns that a GRAND receiver recovers, and by acknowledgments, maintains channel and action posteriors that select the encoding, by Thompson sampling, and order the decoder’s guesses. We analyzed BBE on a two-state Markov channel, where the query order and the posterior update have closed forms and the BLER has a closed form under an independent false-hit model. With these we showed that interleaving helps a memoryless decoder more than one that has learned the bursts, so that the best encoding can change as the decoder learns, and that BBE’s regret is bounded when every frame reveals its noise. In simulation all BBE variants learn the bursts, and BBE-ACK and BBE-hybrid, whose selection uses acknowledgments, interleave early and learn faster when the receiver starts confident that the channel is memoryless.

There are several possible directions for future work. A regret bound under decoded feedback would build on the bias bound of Section V-B. Selection rules that value the burst statistics an interleaved frame returns, such as information-

directed sampling [13], can make the exploration explicit. Finally, the same loop applies to larger action sets, other channel families such as the multi-interferer channel of [3], and decoder parameters such as the query budget.

AI USE STATEMENT

The author has used Claude (Anthropic) for writing code and simulations, for help with the analysis and proofs, and for paper writing, editing, proofreading and formatting.

REFERENCES

- [1] W. An, M. Médard, and K. R. Duffy, “Keep the bursts and ditch the interleavers,” *IEEE Trans. Commun.*, vol. 70, no. 6, pp. 3655–3667, 2022.
- [2] K. R. Duffy, J. Li, and M. Médard, “Capacity-achieving guessing random additive noise decoding,” *IEEE Trans. Inf. Theory*, vol. 65, no. 7, pp. 4023–4040, 2019.
- [3] B. Krishnamachari, “Closed-loop Bayesian bandit encoder with GRAND receiver for a bursty interference channel,” unpublished manuscript, arXiv:2607.15404, 2026.
- [4] B. Krishnamachari, “Scaling Bayesian bandit encoding with shared learning,” unpublished manuscript, arXiv:2609.06293, 2026.
- [5] B. Krishnamachari, “Bayesian bandit encoding: Analysis on a two-state Markov channel (extended version with proofs),” 2026. [Online]. Available: <https://github.com/anrgusc/bbe-2smc>
- [6] A. Solomon, K. R. Duffy, and M. Médard, “Soft maximum likelihood decoding using GRAND,” in *Proc. IEEE ICC*, 2020.
- [7] K. R. Duffy, W. An, and M. Médard, “Ordered reliability bits guessing random additive noise decoding,” *IEEE Trans. Signal Process.*, vol. 70, pp. 4528–4542, 2022.
- [8] H. K. Miyamoto and S. Yang, “Universal decoding over finite-state additive channels via noise guessing,” arXiv:2501.12971, 2025.
- [9] R. Combes, J. Ok, A. Proutière, D. Yun, and Y. Yi, “Optimal rate sampling in 802.11 systems: Theory, design, and implementation,” *IEEE Trans. Mobile Comput.*, vol. 18, no. 5, pp. 1145–1158, 2019.
- [10] H. Qi, Z. Hu, X. Wen, and Z. Lu, “Rate adaptation with Thompson sampling in 802.11ac WLAN,” *IEEE Commun. Lett.*, vol. 23, no. 10, pp. 1888–1892, 2019.
- [11] T. Lattimore and R. Munos, “Bounded regret for finite-armed structured bandits,” in *Proc. NeurIPS*, 2014.
- [12] O. Atan, C. Tekin, and M. van der Schaar, “Global bandits,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 5798–5811, 2018.
- [13] D. Russo and B. Van Roy, “Learning to optimize via information-directed sampling,” *Oper. Res.*, vol. 66, no. 1, pp. 230–252, 2018.